STA2002: Probability and Statistics II
Maximum Likelihood Estimation

Statistical Model

for this course ⇓ in general, we can propose/construct a model form using mathematical tools

▷ focus on "parameter" in a model

(believe true value exist)

▷ estimation of parameters (based on observation)

point estimation is random

interval ~. CI. is random

Instructor: Bojun Lu

School of Data Science
CUHK(SZ)

* Model

H_0 : about parameter

or about "whether model is acceptable."

Problem : model selection

hint : H_0 v.s. H_1.

H_0 : a model denoted as M_0 & H_1 : all other alternative.

also we can : H_0 : X & Y are independent.

February 15, 2023

Introduction

- In this lecture we will learn the maximum likelihood estimation method, which is one of the most common methods about estimating parameters in statistical models.
- **Suggested reading**: Chapter 6.4 in the book.

Introduction (Cont'd)

- Suppose that we have a random sample $X_1, X_2, \ldots, X_n$, i.i.d., which has a common distribution with an unknown parameter denoted as θ and probability mass function (pmf) or probability density function (pdf) $f(x;\theta)$.
- θ takes value in a set, denoted as Ω, which is called the **parameter space**.
- As an example, we can recall the exponential distribution with pdf

$$f(x;\theta) = (1/\theta)e^{-x/\theta}, 0 < x < \infty,$$

and

$$\theta \in \Omega, \quad \Omega := \{\theta;\ 0 < \theta < \infty\}.$$

Introduction (Cont'd)

- The observed values of $X_1, X_2, \ldots, X_n$ are denoted as $x_1, x_2, \ldots, x_n$.
- We now try to estimate θ by using a statistic $u(X_1, X_2, \ldots, X_n)$, which is also called an **estimator** of θ.
- The value of $u(x_1, x_2, \ldots, x_n)$ is said to be an **estimate** of θ.
- Since $u(X_1, X_2, \ldots, X_n)$ is a single value estimate of θ, it is called a **point estimator**.

Two Definitions

- Next, we introduce two important definitions.

Definition 1 (likelihood function)

The likelihood function $L : \Omega \to [0, \infty)$ is the joint density or joint mass function of the observations. L is treated as a function of the parameter θ, that is,

$$L(\theta) := \prod_{i=1}^{n} f(x_i; \theta).$$

The samples denoted as $X_1, X_2 \ldots X_n$ which are Random

assume a distribution with parameters

$L = \pi f(x_i; \theta)$

more than 1.

already realized. (now constant.)

$\therefore$ Only θ unknow.

$\therefore$ likelihood is aways func. in terms of θ.

The log-likelihood function is defined by

$$l(\theta) := \ln L(\theta).$$

Two Definitions

Definition 2 (maximum likelihood estimator)

The maximum likelihood estimator (mle), denoted by $\hat{\theta}$, is the value of θ that maximizes $L(\theta)$.

Note:

- To derive the expression of mle, it is often easier to maximize $l(\theta)$ rather than to maximize $L(\theta)$.
- Since the natural logarithm function is a strictly increasing function, the resulting maximizer $\hat{\theta}$ from maximizing $l(\theta)$ or $L(\theta)$ is the same.

Example 1: Bernoulli(p)

Example 3

Suppose that $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$ *with pmf given by*

$$f(x; p) = p^x (1-p)^{1-x}, \quad x = 0, 1, \quad 0 \le p \le 1.$$

The parameter space is

$$\Omega = \{p;\ 0 \le p \le 1\}.$$

The likelihood function is

figure out the likelihood function is "key" part

$$L(p) = \prod_{i=1}^{n} f(x_i; p) = p^{\sum_{i=1}^n x_i} (1-p)^{n - \sum_{i=1}^n x_i}.$$

$\max_\theta L(\theta)$ s.t $\theta \in \Omega$

sol: $\ln[L(p)] = \ln\left(p^{\Sigma x_i}(1-p)^{n-\Sigma x_i}\right) = \Sigma x_i \ln p + (n - \Sigma x_i)\ln(1-p)$

$\frac{\partial \ln[L(p)]}{\partial p} = \Sigma x_i \cdot \frac{1}{p} - (n - \Sigma x_i) \cdot \frac{1}{1-p} \overset{\text{set}}{=} 0$

$\Rightarrow p = \frac{\Sigma x_i}{n} = \bar{X}$

Task: To derive the mle of p.

Solution: $\hat{p} = \overline{X}$ with $\overline{X} = \sum_{i=1}^n X_i / n$.

Example 2: Exponential(θ)

Example 4

Suppose that $X_1, \ldots X_n$ are i.i.d. and follow an exponentially distributed with parameter θ and pdf as below,

$$f(x;\theta) = \frac{1}{\theta} e^{-x/\theta}, \quad 0 < x < \infty, \quad \theta \in \Omega = \{\theta : 0 < \theta < \infty\}.$$

The likelihood function is

$$L(\theta) = \prod_{i=1}^{n} f(x_i;\theta) = \prod_{i=1}^{n} \frac{1}{\theta} e^{-x_i/\theta} = \frac{1}{\theta^n} e^{-\sum_{i=1}^{n} x_i/\theta}.$$

$\frac{\partial l(\theta)}{\partial(\theta)} = \frac{-n}{\theta} + \frac{1}{\theta^2}\sum x_i \Rightarrow \theta = \frac{\sum x_i}{n} = \bar{x}$

The log-likelihood function is

$$l(\theta) = -n \ln \theta - \frac{1}{\theta} \sum_{i=1}^{n} x_i, \quad 0 < \theta < \infty.$$

Task: To derive the mle of θ. Solution: $\hat{\theta} = \overline{X}$ with $\overline{X} = \sum_{i=1}^{n} X_i/n$.

Example 3: Geometric(p)

Example 5

Suppose that $X_1, \ldots X_n$ follow a geometric distribution with success probability p and pmf given as below,

$$f(x;p) = (1-p)^{x-1}p, x = 1,2,3,\ldots, \quad p \in \Omega = \{p : 0 < p < 1\}.$$

The likelihood function is

$$L(p) = \prod_{i=1}^{n} f(x_i;p) = \prod_{i=1}^{n}(1-p)^{x_i-1}p = p^n(1-p)^{\sum_{i=1}^n x_i - n}.$$

The log-likelihood function is

$$\frac{\partial l(p)}{\partial p} = \frac{n}{p} - \frac{\sum x_i - n}{1-p} \Rightarrow p = \frac{\sum x_i}{n} = \bar{x}$$

$$l(p) = n \ln p + \left(\sum_{i=1}^{n} x_i - n\right)\ln(1-p).$$

Task: To derive the mle of p. Solution: $\hat{\theta} = 1/\overline{X}$ with $\overline{X} = \sum_{i=1}^n X_i/n$.

Example 4: Uniform$[0, \theta]$

Example 6

Suppose that $X_1, \ldots X_n$ are i.i.d. and follow a uniformly distributed on the interval $[0, \theta]$ with pdf given as below,

$$f(x;\theta) = \begin{cases} \frac{1}{\theta}, & \text{for } 0 \le x \le \theta, \\ 0, & \text{otherwise}, \end{cases} \quad , \quad \theta \in \Omega = \{\theta : 0 < \theta < \infty\}.$$

p.d.f.

Task: To derive the mle of θ.

$L(\theta) = \prod_{i=1}^{n} \frac{1}{\theta} = \theta^{-n}$ $x_1, x_2 \ldots x_n \in (0, \infty)$ ↓ with θ

n is sample size

$\max_{\theta} L(\theta)$ s.t $\theta \ge x_1, x_2 \ldots x_n \Rightarrow \hat{\theta} \triangleq \max\{x_1, x_2 \ldots x_n\}$

$\because L(\theta) \uparrow$ when $\theta \downarrow$

Solution: $\hat{\theta} = X_{(n)}$ with

$$X_{(n)} := \max\{X_1, \ldots, X_n\}.$$

Why Maximizing the Likelihood Function

Assumption 7 (Regularity Conditions)

Regularity conditions (R0)-(R2) are given by

(R0) *The pdfs are distinct; i.e.,* $\theta \neq \theta' \Rightarrow f(x_i;\theta) \neq f(x_i;\theta')$.

(R1) *The pdfs have common support for all* θ.

(R2) *The point* θ_0 *is an interior point in* Ω.

Theorem 8

Let θ_0 *be the true parameter, under assumptions (R0) and (R1), we have*

$$\lim_{n\to\infty} P_{\theta_0}[L(\theta_0;x) > L(\theta;x)] = 1, \quad \text{for all } \theta \neq \theta_0.$$

Note: This theorem is not required in this course. For the students who may have interest. To better understand the concept of mle.

MLE Again

- Theorem 8 implies that asymptotically (渐近的) the likelihood function L is maximized at the true value θ_0.
- So in considering estimates of θ_0, it is natural to consider the value of θ which maximizes the likelihood.
- Here comes the definition of Maximum Likelihood Estimator (mle).

Definition 9 (Maximum Likelihood Estimator)

We say that $\hat{\theta} = \hat{\theta}(X_1, \ldots, X_n)$ is a maximum likelihood estimator (mle) of θ if

$$\hat{\theta} = \textit{Argmax } L(\theta; X_1, \ldots, X_n), \tag{1}$$

where the notation Argmax means that $L(\theta; X_1, \ldots, X_n)$ achieves its maximum value at $\hat{\theta}$.